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Mysuru – 570 008

Project Report

Course : Linear Intergeted Circuit

Topic: OpAmp

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Characteristics and parameters of Op-Amp

To study and measure the characteristics and parameters of an Op-amp (IC-741):

- (a) Input bias current at inverting terminal
- (b) Input bias current at non-inverting terminal
- (c) Input offset current
- (d) Input offset voltage (e) Slew rate

Introduction

An operational amplifier is a direct coupled high gain amplifier usually consisting of one or more differential amplifiers and followed by a level translator and an output stage. The output stage is generally a push-pull or push-pull complementary-symmetry pair. An operational amplifier is available as a single integrated circuit package.

The operational amplifier is a versatile device that can be used to amplify dc as well as ac input signals and was originally designed for performing mathematical operations such as addition, subtraction, multiplication and integration. Thus, the word operational amplifier stems for its original use for these mathematical operations and is abbreviated to op-amp. With the addition of suitable external feedback components, the modern day op-amp can be used for a variety of applications, such as ac and dc signal amplification, active filters, oscillators, comparators, regulators and others. Since an op-amp is a multistage amplifier, it can be represented by a block diagram as shown in Fig. 1.

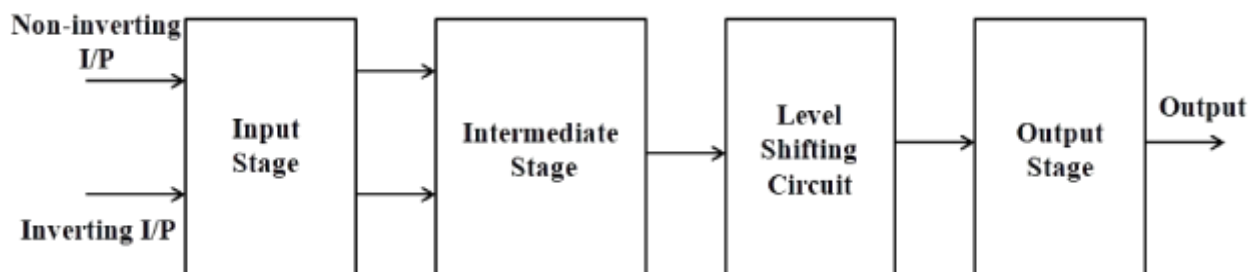


Fig1:Block diagram of op-amp

- The input stage of an operational amplifier is typically a dual-input differential amplifier with balanced output. This stage is responsible for providing a large portion of the overall voltage gain and also determines

the **input resistance** of the op-amp. The signal from this stage is then fed to the intermediate stage, which is usually another differential amplifier. In many amplifier designs, this stage operates with dual inputs but produces a single-ended output.

Since the amplifier stages are directly coupled, the DC voltage level at the output of the intermediate stage becomes higher than the ground potential. To correct this, a level shifting (level translator) circuit is introduced after the intermediate stage. Its purpose is to reduce the DC voltage level and bring it closer to zero volts with respect to ground.

The final stage of the op-amp is generally a push–pull complementary output stage. This stage improves the output voltage swing and increases the current driving capability of the amplifier. Additionally, a properly designed output stage ensures that the op-amp has a low output resistance, allowing it to effectively drive external loads.

† Schematic symbol

Fig. 2 shows the schematic symbol of an op-amp

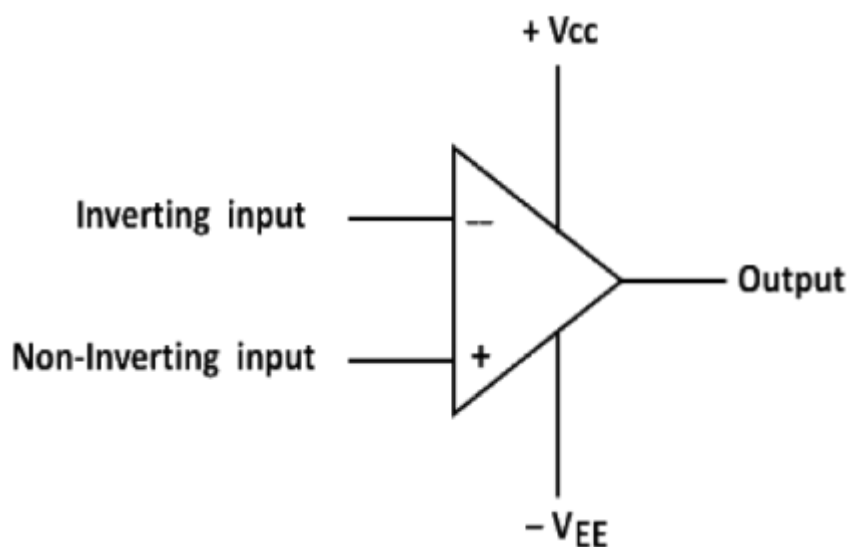


Fig-2: Schematic symbol of op-amp

The input differential amplifier stage of the op-amp is designed to be operated in the differential mode, the differential inputs are designated by the (+) and (-) notations. The (+) input is the non-inverting input. An ac signal (or dc voltage) applied to this input produces an in-phase (or same polarity) signal at the output. On the other hand, the (-) input is the inverting input because an ac signal (or dc voltage) applied to this input produces a 180° out of phase (or opposite polarity) signal at the output.

Op-amp IC-741 pin configuration

The pin configuration of IC-741 general purpose operational amplifier is shown in Fig. 6. The reference point of the chip is the notch at the top. The pins are numbered in a counter clockwise direction, starting at the notch.

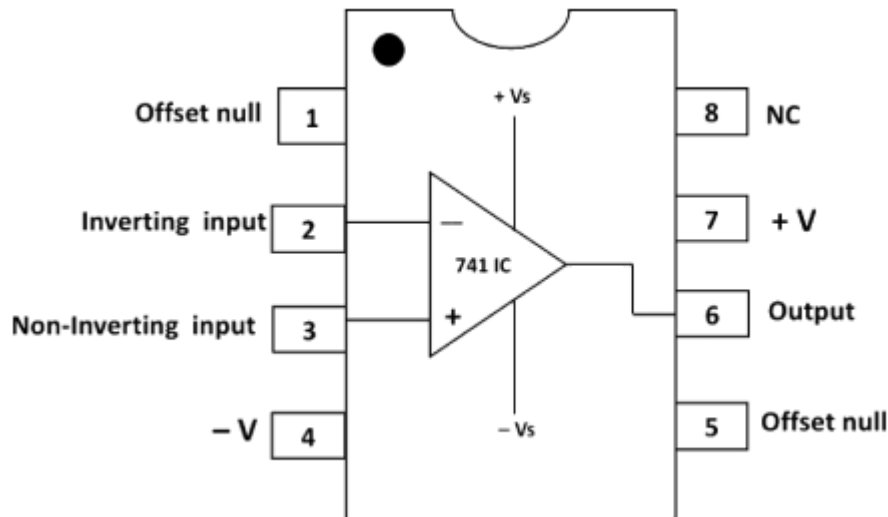


Fig-3: Pin diagram of op-amp IC-741

- **Pin 1 and Pin 5** are known as offset null pins. These pins are used to correct small output errors caused by internal imbalances in the op-amp. Sometimes the op-amp produces a small output voltage even when no input signal is applied. To minimize this effect, these pins are connected to a variable resistor (potentiometer). By adjusting the potentiometer, the output voltage at Pin 6 can be made approximately zero.
- **Pin 2 and Pin 3** are the **input terminals** of the op-amp. Both inputs influence the output at Pin 6, but they affect the polarity differently. Pin 2 is the inverting input, meaning that a signal applied to this pin produces an output that is **180°** out of phase with the input. Pin 3 is the non-inverting input, and a signal applied here results in an output that has the same polarity as the input signal.
- **Pin 4 and Pin 7** are used to provide the power supply to the op-amp. **Pin 4** is connected either to ground or to a negative supply voltage (typically between -3 V and -18 V), while Pin 7 is connected to the positive supply voltage.
- **Pin 6** is the output terminal of the operational amplifier, where the amplified signal is obtained.
- **Pin 8** is not connected internally and is labeled NC (No Connection).

Characteristics of an ideal op-amp:

An ideal op-amp would exhibit the following electrical characteristics:

- Infinite voltage gain A .
- Infinite input resistance R_i , so that almost any signal source can drive it and there is no loading of the preceding stage.
- Zero output resistance R_o , so that the output can drive an infinite number of other devices.
- Zero output voltage when input voltage is zero.
- Infinite bandwidth, so that any frequency signal from 0 to ∞ Hz can be amplified without attenuation.
- Infinite common-mode rejection ratio (CMRR), so that the output common mode noise voltage is zero.
- Infinite slew rate, so that output voltage changes occur simultaneously with input voltage changes.

Parameters of an op-amp

Parameters of an Operational Amplifier

Operational amplifiers have several important parameters that determine their performance in practical circuits. These parameters help in understanding how closely a real op-amp behaves compared to an ideal op-amp.

1. Input Offset Voltage (V_{io})

Input offset voltage is the small differential voltage that must be applied between the inverting and non-inverting input terminals to make the output voltage equal to zero. In an ideal op-amp this value should be zero, but in practical devices a small offset voltage exists due to internal transistor mismatches.

2. Input Bias Current (I_B)

Input bias current is the small DC current that flows into the input terminals of an operational amplifier. This current is required for the proper operation of the internal transistor circuits. In an ideal op-amp the input bias current is zero, but in practical op-amps a small current flows at both input terminals.

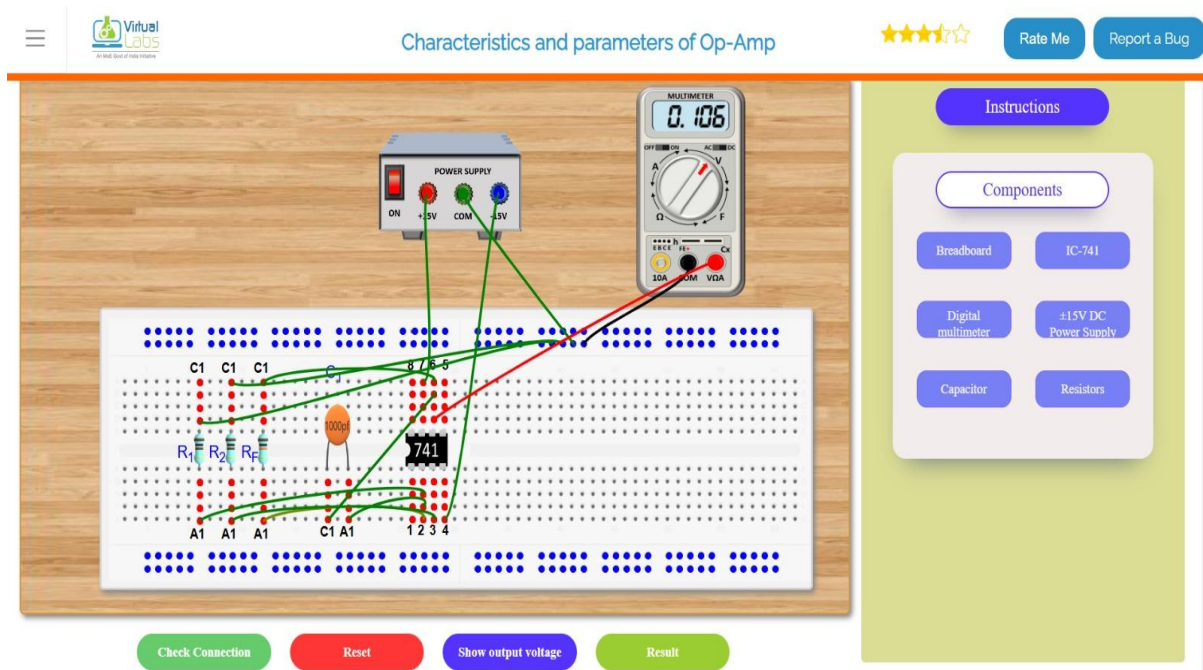
3. Input Offset Current (I_{io})

Input offset current is defined as the difference between the bias currents of the inverting and non-inverting input terminals. Ideally both currents should be equal, but due to manufacturing variations a small difference exists.

4. Slew Rate (SR)

Slew rate is the maximum rate of change of the output voltage with respect to time. It indicates how quickly the output of the op-amp can respond to rapid changes in the input signal. It is usually expressed in volts per microsecond ($V/\mu s$).

1. Input Offset Voltage (V_{io}):



Procedure:

1. Click on the components button to place the component on the table.
2. Make connections as per the circuit diagram or connection table.
3. Click on 'Check Connection' button to check connections. If correct, Click on 'Show output voltage' button to view output (V_o) on DMM.
4. Calculate the input offset voltage, V_{io} using the formula:
$$V_{io} = (V_o - I_{io}R_f)/(1+R_f/R_i)$$
5. Click on 'Result' button and enter the calculated value.

Table 4: Connection table to measure input offset voltage.

S. No.	Source	Target
1	IC741_ pin 07	Supply VCC
2	Multimeter_GND	Supply GND
3	IC741_ pin 04	Supply VEE
4	Resistance (R_f)_terminal A1	IC741_ pin 02
5	Resistance (R_f)_terminal C1	IC741_ pin 06
6	Capacitance (C_1) _terminal A1	IC741_ pin 02
7	Capacitance (C_1) _terminal C1	IC741_ pin 06
8	Resistance (R_2) _terminal A1	IC741_ pin 03
9	Resistance (R_2) _terminal C1	Supply GND
10	Resistance (R_1) _terminal A1	IC741_ pin 02
11	Resistance (R_1) _terminal C1	Supply GND
12	Multimeter_VCC	IC741_ pin 06

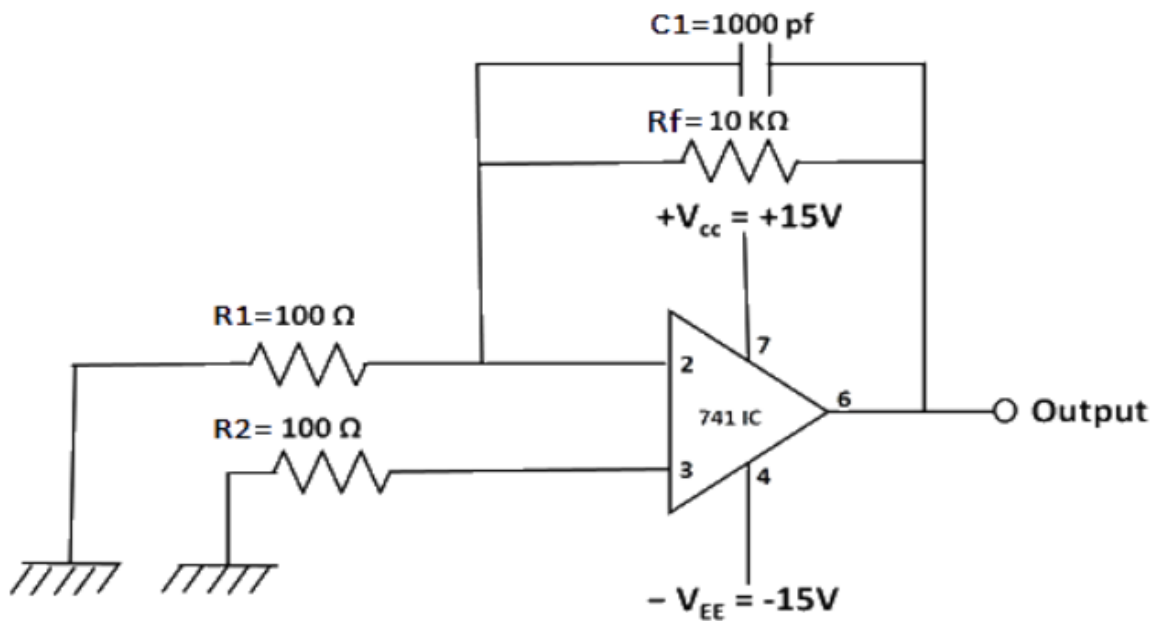


Fig 6: Circuit diagram to measure input offset voltage.

Input offset voltage is the small differential voltage that must be applied between the inverting and non-inverting input terminals of an operational amplifier to make the output voltage equal to zero. In an ideal op-amp, this value should be zero, but in practical devices a small offset voltage exists due to internal transistor mismatches.

In this experiment, the operational amplifier is connected in a high-gain closed-loop configuration. Both input terminals are connected to ground through resistors. Ideally, the output voltage should be zero when no input signal is applied. However, because of the internal imbalance in the op-amp, a small DC voltage appears at the output.

The relationship between the output voltage and the input offset voltage is given by:

$$V_{out} = \left(1 + \frac{R_f}{R_1}\right) V_{io}$$

From this relation, the input offset voltage can be calculated as:

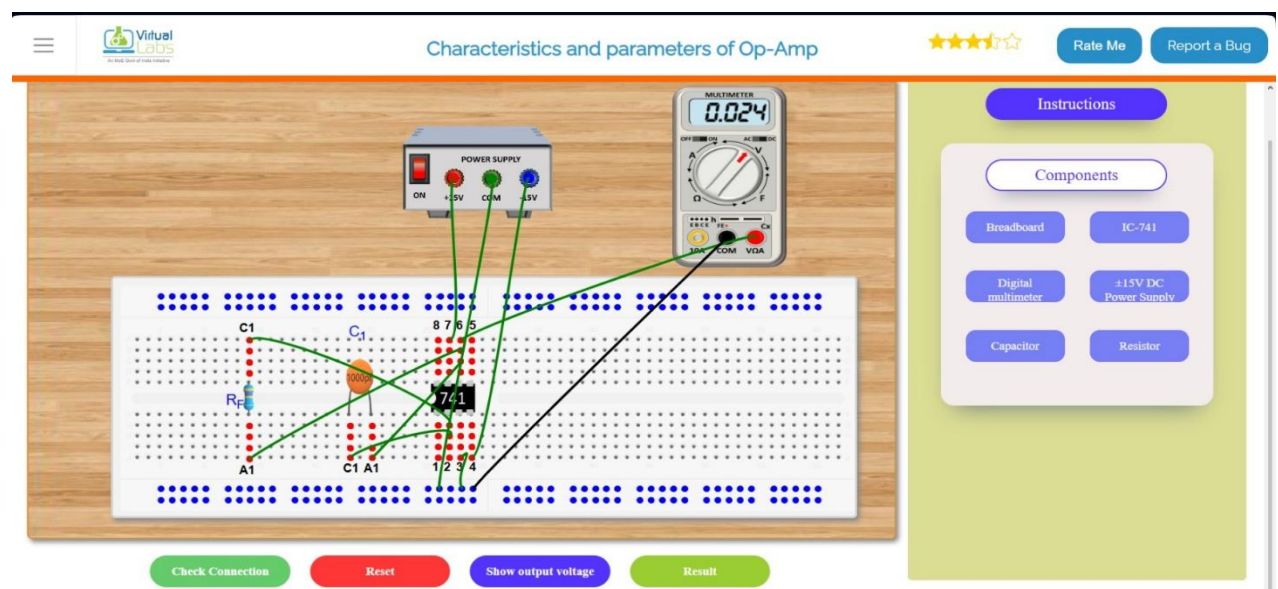
$$V_{io} = \frac{V_{out}}{1 + \frac{R_f}{R_1}}$$

By measuring the output voltage and knowing the values of the feedback resistor and input resistor, the input offset voltage of the operational amplifier can be determined.

The measured input offset voltage of the IC 741 is **1.048 mV**.

2. Input Bias Current (IB)

2.1 Inverting Bias Current:



Procedure:

1. Click on the components button to place the component on the table.
2. Make connections as per the circuit diagram or connection table.
3. Click on 'Check Connection' button to check connections. If correct, Click on 'Show output voltage' button to view output on DMM.
4. Calculate the inverting bias current using the formula:
$$I_{B1} = V_o/R_f$$
5. Click on 'Result' button and enter the calculated value.
6. Click on 'Reset' button and proceed in the same way to calculate the input bias current at non-inverting terminal.

Table 1: Connection table to measure inverting bias current.

S. No.	Source	Target
1	IC741_ pin 07	Supply VCC
2	IC741_ pin 03	Supply GND
3	Multimeter _GND	Supply GND
4	IC741_ pin 04	Supply VEE
5	IC741_ pin 02	Resistance (R_f) _ terminal C1
6	IC741_ pin 06	Resistance (R_f) _ terminal A1
7	IC741_ pin 02	Capacitance (C_1) _ terminal C1
8	IC741_ pin 06	Capacitance (C_1) _ terminal A1
9	Multimeter_VCC	IC741_ pin 06

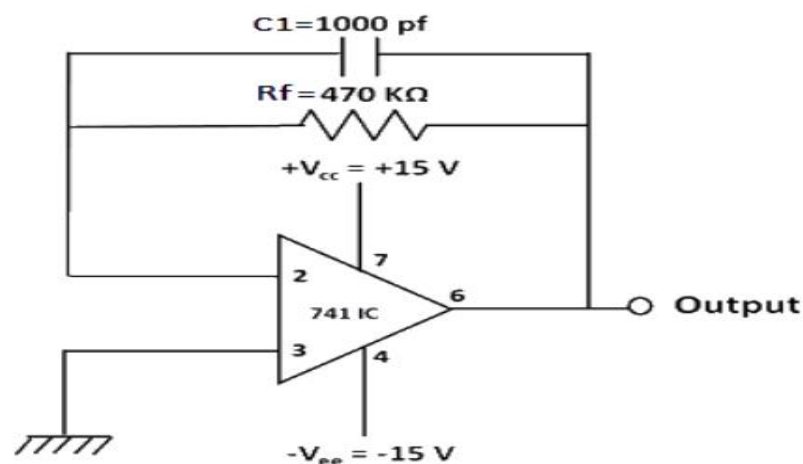


Fig 2: Circuit diagram to measure inverting bias current.

In an ideal operational amplifier, no current flows into the input terminals. However, in a practical op-amp such as the IC 741 Operational Amplifier, a small current flows into the input terminals due to the internal transistor biasing. This current is known as the input bias current.

The current flowing into the inverting terminal is called the inverting input bias current. In this experiment, the inverting bias current is measured by connecting a large feedback resistor between the output and the inverting input, while the non-inverting input is connected to ground. Because the feedback resistance is high, even a very small bias current produces a measurable output voltage. The output voltage is related to the inverting bias current by the equation:

$$V_o = I_{B1} \times R_f$$

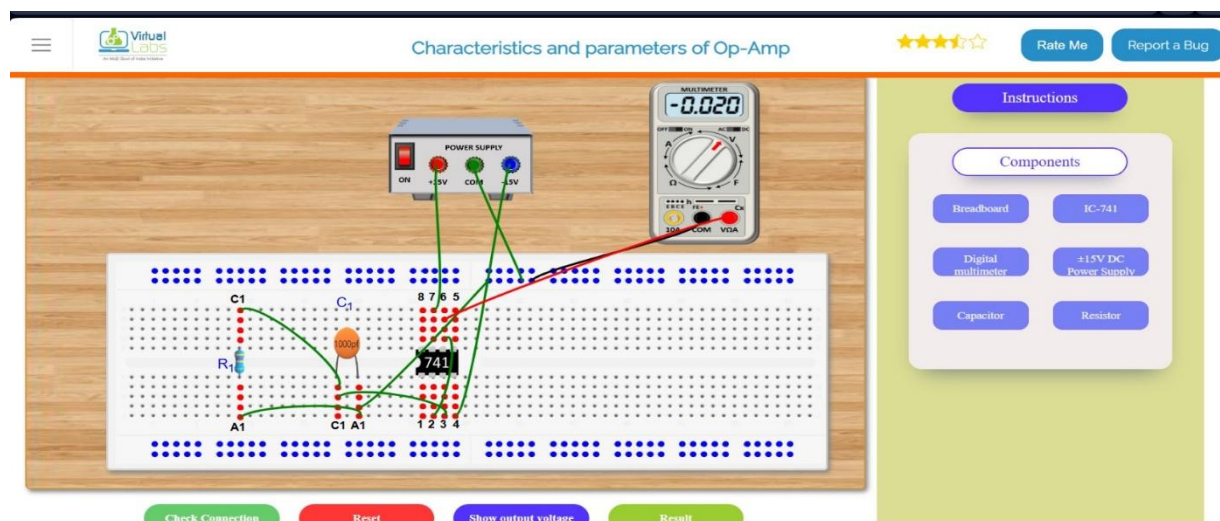
Therefore, the inverting bias current can be calculated as:

$$I_{B1} = \frac{V_o}{R_f}$$

By measuring the output voltage and knowing the value of the feedback resistor, the inverting bias current of the operational amplifier can be determined.

The measured inverting input bias current of the IC 741 is **0.0510 μ A**.

2.2 Non-Inverting Bias Current:



Procedure:

1. Click on the components button to place the component on the table.
2. Make connections as per the circuit diagram or connection table.
3. Click on 'Check Connection' button to check connections. If correct, Click on 'Show output voltage' button to view output on DMM.

4. Calculate the non-inverting bias current using the formula:

$$I_{B2} = V_o/R_1$$

5. Click on 'Result' button and enter the calculated value.

6. The input bias current I_B hence, can be calculated using the formula:

$$I_B = (I_{B1} + I_{B2})/2$$

where, I_{B1} has to be calculated from previous part.

Table 2: Connection table to measure non-inverting bias current.

S. No.	Source	Target
1	IC741_ pin 07	Supply VCC
2	Multimeter_GND	Supply GND
3	IC741_ pin 06	IC741_ pin 02
4	IC741_ pin 04	Supply VEE
5	IC741_ pin 03	Resistance (R_1) _terminal C1
6	Supply GND	Resistance (R_1) _terminal A1
7	IC741_ pin 03	Capacitance (C_1) _terminal C1
8	Supply GND	Capacitance (C_1) _terminal A1
9	Multimeter_VCC	IC741_ pin 06

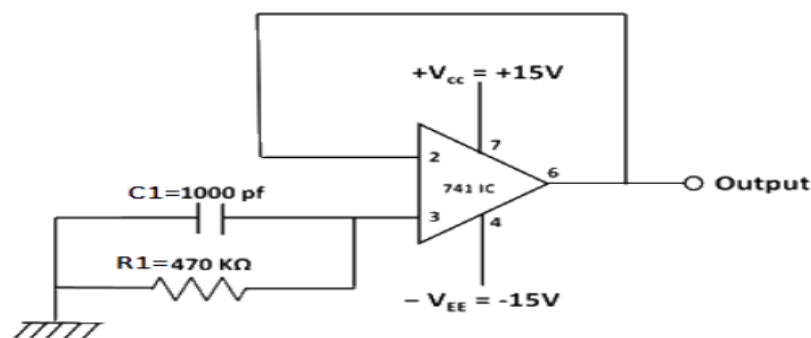


Fig 3: Circuit diagram to measure non-inverting bias current.

In a practical operational amplifier such as the IC 741 Operational Amplifier, a small amount of current flows into both input terminals due to the internal transistor biasing. The current flowing into the non-inverting input terminal is called the non-inverting bias current.

In this experiment, the inverting input is connected through a feedback network, while a resistor is connected to the non-inverting terminal. Because of the bias current flowing into this terminal, a small voltage drop develops across the resistor. This voltage appears at the output of the op-amp.

The relationship between the output voltage and the non-inverting bias current is given by:

$$I_{B2} = \frac{V_o}{R_1}$$

Where

I_{B2} = Non-inverting input bias current

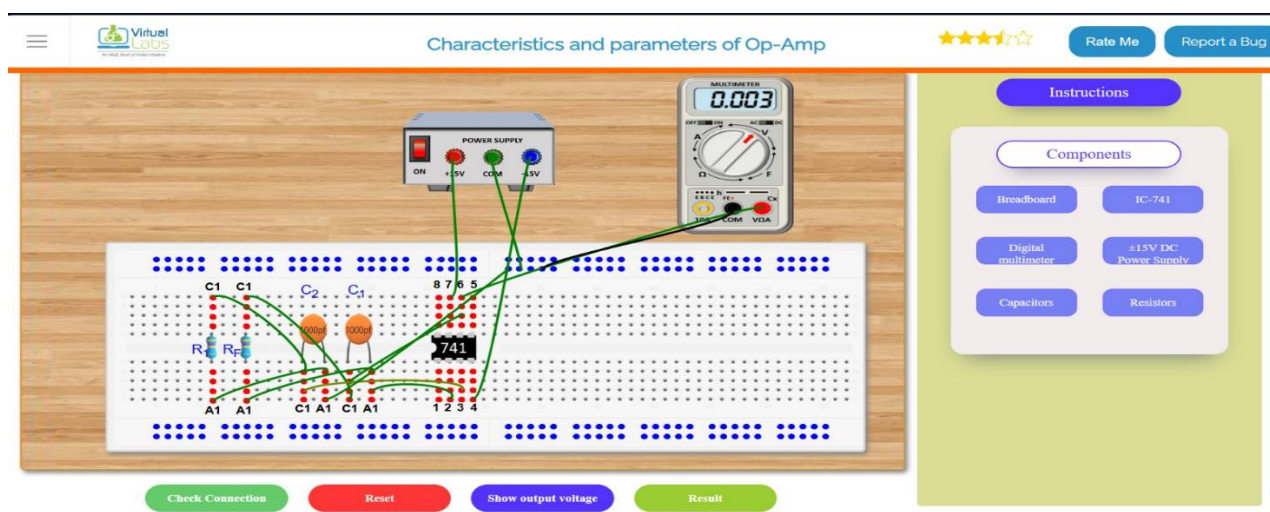
V_o = Output voltage

R_1 = Resistance connected to the non-inverting input

By measuring the output voltage and knowing the value of the resistor, the non-inverting bias current of the operational amplifier can be calculated.

The measured non-inverting input bias current of the IC 741 operational amplifier is **-0.0425 μ A**.

2. Input Offset Current (I_{io}):



Procedure:

1. Click on the components button to place the component on the table.
2. Make connections as per the circuit diagram or connection table.
3. Click on 'Check Connection' button to check connections. If correct, Click on 'Show output voltage' button to view output on DMM.
4. Calculate the input offset current using the formula:

$$I_{io} = V_o / R_f$$
5. Click on 'Result' button and enter the calculated value.

Table 3: Connection table to measure input offset current.

S. No.	Source	Target
1	IC741_ pin 07	Supply VCC
2	Multimeter_GND	Supply GND
3	IC741_ pin 04	Supply VEE
4	Resistance (R_f)_ terminal A1	IC741_ pin 02
5	Resistance (R_f)_ terminal C1	IC741_ pin 06
6	Capacitance (C_1) _terminal A1	IC741_ pin 02
7	Capacitance (C_1) _terminal C1	IC741_ pin 06
8	Resistance (R_1) _ terminal A1	IC741_ pin 03
9	Resistance (R_1) _ terminal C1	Supply GND
10	Capacitance (C_2) _terminal A1	IC741_ pin 03
11	Capacitance (C_2) _terminal C1	Supply GND
12	Multimeter_VCC	IC741_ pin 06

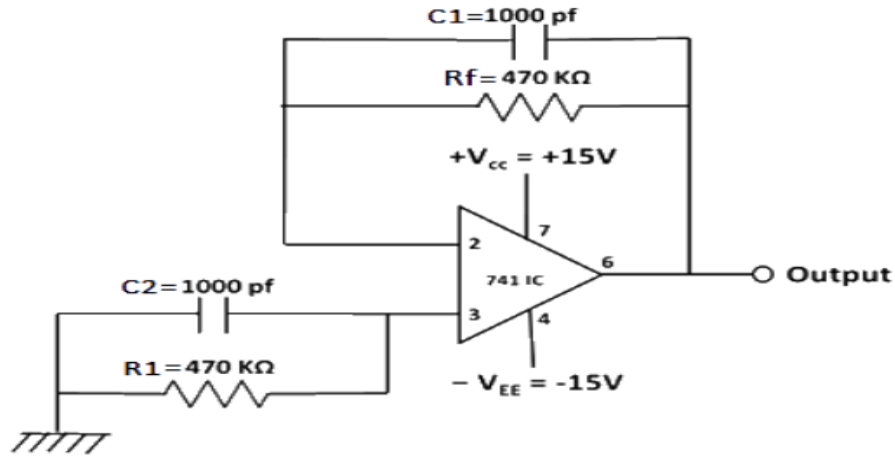


Fig 4: Circuit diagram to measure input offset current.

In an ideal operational amplifier, the currents flowing into the inverting and non-inverting input terminals are equal and ideally zero. However, in a practical operational amplifier such as the IC 741 Operational Amplifier, small currents flow into the input terminals due to the internal transistor biasing.

The input offset current is defined as the difference between the bias currents of the inverting and non-inverting input terminals. Because the internal transistors are not perfectly matched, these two currents are not exactly equal, which results in a small difference known as the offset current.

In this experiment, the operational amplifier is connected with a high feedback resistance. Even a very small offset current flowing through this resistor produces a measurable output voltage.

The relationship between the output voltage and the input offset current is given by:

$$V_o = I_{io} \times R_f$$

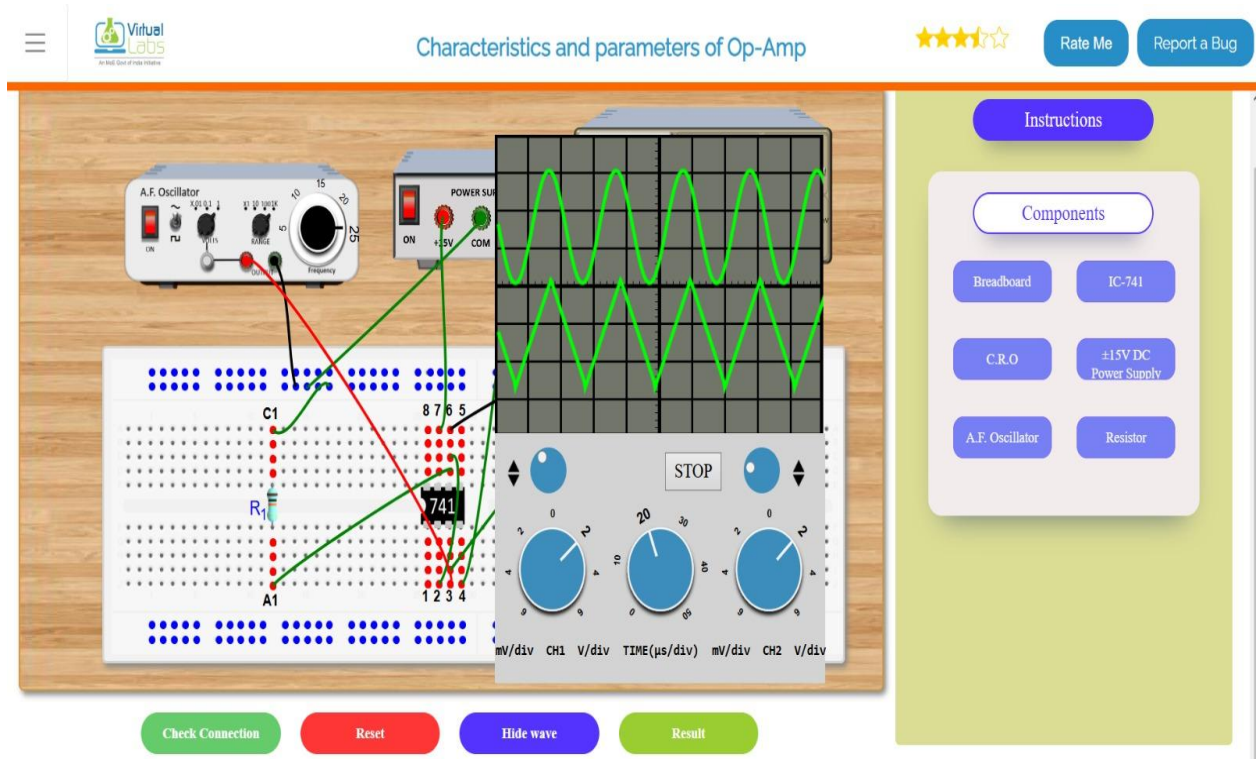
Therefore, the input offset current can be calculated as:

$$I_{io} = \frac{V_o}{R_f}$$

By measuring the output voltage and knowing the value of the feedback resistor, the input offset current of the operational amplifier can be determined.

The measured input offset current of the IC 741 is **0.0063 μ A**

4. Slew Rate:



Procedure:

1. Click on the components button to place the component on the table.
2. Make connections as per the circuit diagram or connection table.
3. The input signal* has to be taken from the red terminal of A.F. Oscillator while the other terminal has to be grounded.
4. Also, feed the input signal to channel CH1 of the C.R.O and the output from Op-Amp IC must be fed to the channel CH2.

5. Click on 'Check Connection' button to check connections. If correct, Click on 'Show output voltage' button to view output.

6. Increase the frequency of input signal using the dial** on A.F. Oscillator until output wave becomes triangular. This is the required frequency $f_{\max}$ in KHz.

7. Calculate the slew rate using the formula:

$$\text{S.R.} = 2\pi f_{\max} V_m / 10^6 \text{ V}/\mu\text{s}$$

8. Click on 'Result' button and enter the calculated value.

*Amplitude of input signal ' V_m ' is 3V.

** dial can be controlled through both mouse and keyboard.

Table 5: Connection table to measure slew rate.

S. No	Source	Target
1	IC741_ pin 07	Supply VCC
2	IC741_ pin 04	Supply VEE
3	Generator_ terminal C	Supply GND
4	Resistance (R_1) _terminal A1	IC741_ pin 06
5	Resistance (R_1) _terminal C1	Supply GND
6	IC741_ pin 06	IC741_ pin 02
7	Generator_ terminal A	IC741_ pin 03
8	CRO_ terminal C	IC741_ pin 06
9	Generator_ terminal A	CRO_ terminal A

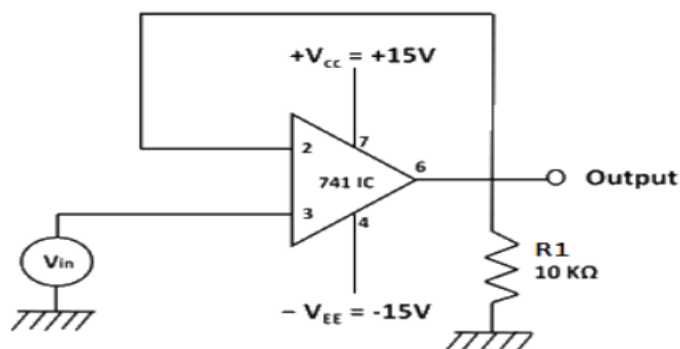


Fig 8: Circuit diagram to measure slew rate.

Slew rate is defined as the maximum rate of change of the output voltage with respect to time in an operational amplifier. It indicates how fast the output of the amplifier can respond to rapid changes in the input signal. Slew rate is usually expressed in volts per microsecond ($V/\mu s$).

In an ideal operational amplifier, the output voltage would change instantly with the input signal. However, in practical operational amplifiers such as the IC 741 Operational Amplifier, the rate at which the output voltage can change is limited due to internal capacitances and circuit design.

In this experiment, a high-frequency sinusoidal signal is applied to the input of the op-amp. As the frequency increases, the output waveform begins to distort and may appear triangular in shape. This distortion occurs because the output cannot change fast enough to follow the input signal.

Two waveforms are visible:

- Upper waveform: Input signal (approximately a sinusoidal wave from the AF oscillator).
- Lower waveform: Output signal from the op-amp.

What we can conclude from the graph

1. Output distortion is present
The output waveform looks triangular instead of sinusoidal, which indicates that the op-amp is not able to follow the input signal perfectly.
2. Slew Rate Limitation
In a 741 op-amp, the rate at which the output voltage can change is limited (called slew rate).
When the input frequency or amplitude is high, the op-amp cannot change fast enough, so the sine wave gets converted into a triangular waveform.
3. Amplification is occurring
The output waveform has larger amplitude compared to the input, indicating that the op-amp is providing gain.
4. Possible phase shift
Depending on the configuration (inverting/non-inverting), the output may also show a phase difference relative to the input.

The measured slew rate of the IC 741 is **0.471 $V/\mu s$**

Inference:

From the experiment, the important parameters of the operational amplifier such as input offset voltage, input bias current, input offset current, and slew rate were measured successfully. The results show that a practical operational amplifier has small offset voltages and bias currents due to internal transistor mismatches and biasing requirements. The measurements also confirm that the slew rate limits the maximum speed at which the output voltage can change in response to the input signal. These observations indicate that practical op-amps slightly deviate from ideal characteristics.

Conclusion:

The experiment was performed to study the characteristics and parameters of the IC 741 Operational Amplifier. The values of input offset voltage, input offset current, inverting and non-inverting bias currents, and slew rate were determined using the given circuits. The results obtained are within the expected range for the IC 741 operational amplifier. This experiment helps in understanding the practical behavior and limitations of operational amplifiers used in electronic circuits.